NPSHr 40000hours – Vlaming formula method and calculation

Index

1. Purpose 3

2. Scope 3

3. Definitions 3

4. General Procedure - Description 4

5. Methodology, analysis and interpretation of results 5

6. Limitations 14

7. Acceptance Criteria 14

8. References 15

1. Purpose

This document establishes the criteria to calculate the Net Positive Suction Head required to ensure 40000 hours as a reasonable life of an impeller against cavitation damage by using Vlaming [1] and Schiavello [2] formulations. If NPSH available exceeds this value (which tends to be higher than the standard NPSH required) the Manufacturer effectively guarantees no loss of performance or parts replacement for 40,000 hours (about 4.5 years) of continuous operation.

2. Scope

This procedure is applicable to both single and double suction first stage impeller, single- or multi-stage pumps.

3. Definitions

b blade angle of the impeller leading edge deg

A Impeller inlet area m^2 or in^2

c flow absolute velocity at the inlet of the impeller tip at shockless condition m/s or ft/s

Ca Correction factor for specific speed

Cb Correction factor for gas to liquid ratio

c1 flow absolute velocity at the impeller inlet at rated condition m/s or ft/s

D1 inlet diameter of the impeller at tip m or in

K1 friction coefficient for the casing

K2 friction coefficient for the impeller

L length of the shaped leading edge m or in

n rotating speed of the pump rpm

Nq (Ns) specific speed of the pump rpm*(Q/impeller suctions)$^{0.5}$/(H/stages)$^{0.75}$

NPSHRse required net positive suction head for shockless condition m or in

QRTD rated flowrate of the pump m^3/h or GPM

QBEP flowrate of the pump at bep m^3/h or GPM

HBEP head of the pump at bep m3/h or ft

QSE flowrate of the pump at shockless condition

R gas to liquid volume fraction

S thickness of the impeller leading edge mm or in

U peripheral velocity at impeller D1 m/s or ft/s

V specific volume of the vapour fluid m^3/kg or ft^3/pd

Wserelative velocity on the impeller tip at shockless condition

4. General Procedure - Description

The evaluation of pump NPSH40000 is calculated using the “Mod.421” spreadsheet.

This method considers of the following pump features:

- Single/double suction 1st stage impeller
- Rotational speed
- Head per stage at rated and bep condition.
- Pump flow rate at rated bep and shockless condition
- Volumetric efficiency

- NPSH available
- Gas to liquid ratio R. The suggested value can be extracted by selecting from a list of special fluid or by setting the proper water temperature. The calculated value of R must be copied in the blank cell.
- Geometry of the impeller leading edge in term of impeller eye diameter, inlet width of the meridional channel, hub diameter of the impeller
- Blade angles of the impeller at the leading edge.
- Shape and thickness of the leading edge

5. Methodology, analysis and interpretation of results

The calculation on NPSH40000hr is based on the Vlaming [1] formulation below:

$$\frac{K1 + K2}{2 * g} * U^2 * \left(\tan(\beta)^2 + \frac{K1}{K1 + K2}\right) * C_a * C_b$$

Eq.1

- K1 is the friction coefficient for the casing. Typical value is 1.2

- K2 is friction coefficient for the impeller. Is strongly influenced by the shaping of the leading edge. It's calculated using the graph in Figure 1

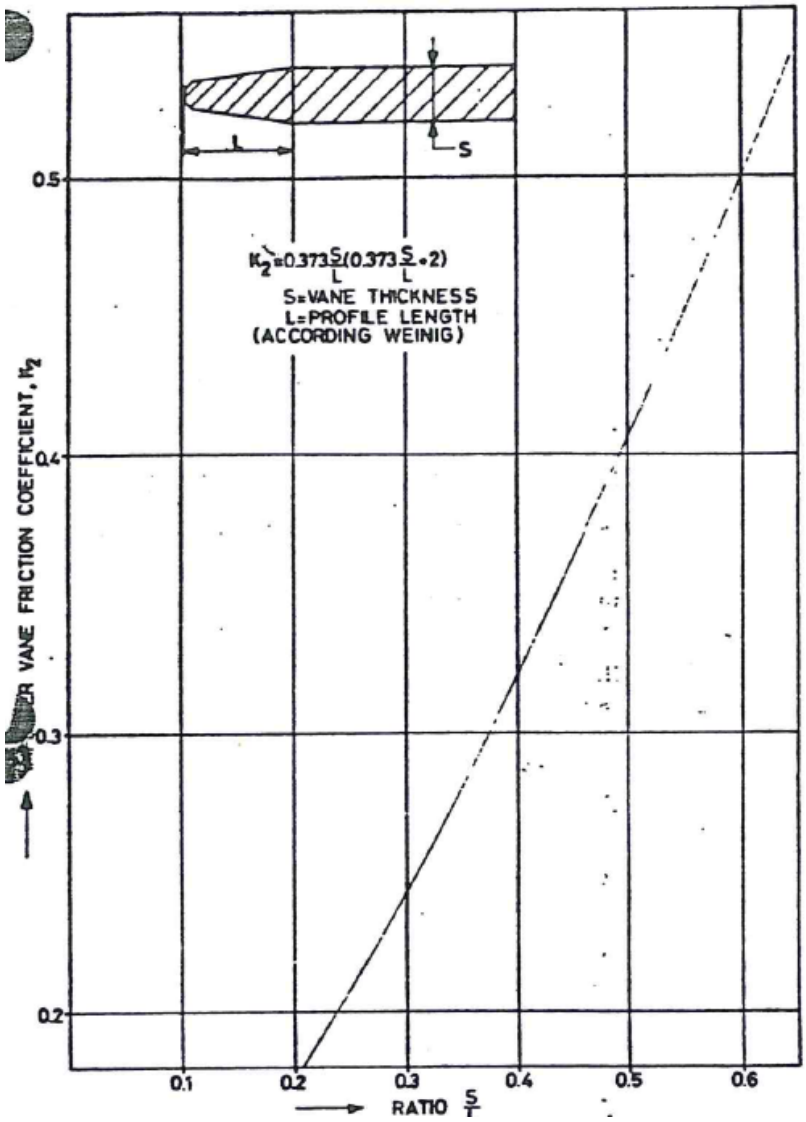

Figure 1. k2 evaluation as function of Leading edge tapering (figure 2 of [1])

Ca is the correction factor for specific speed. It's calculated using the graph in Figure 2

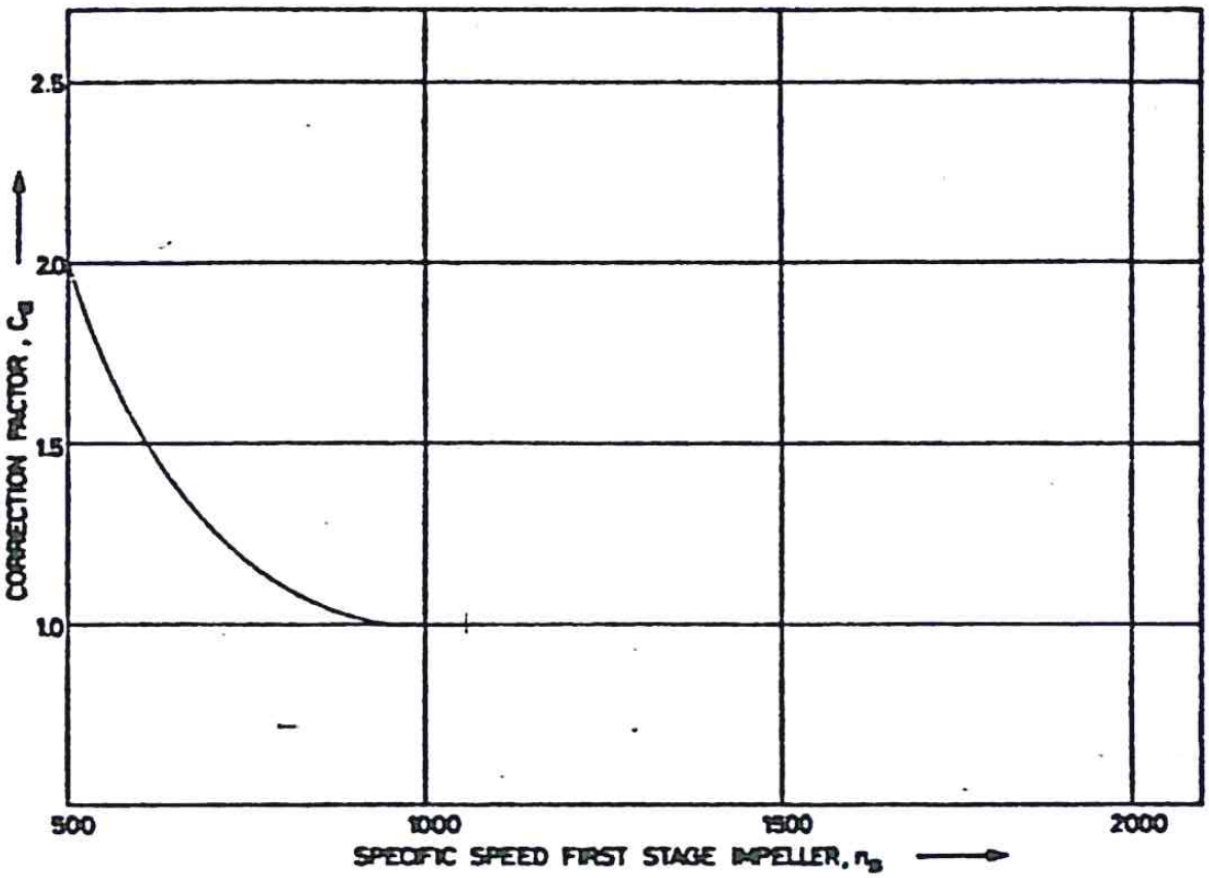

Figure 2. Ca evaluation as function of specific speed (figure 3 of [1])

Cb is the correction factor for the das to liquid ratio and is calculated by using the graph in Figure 3

Cb is calculated for some special fluids like

- crude oil or nitrogen at -320°F (-196°C)
- butane at 30 °F (-1°C)
- prophane at -40°F (-40°C)
- methane at 258°F (125°C)
- oxygen at -300°F (-184°C)

or also for water at different temperatures.

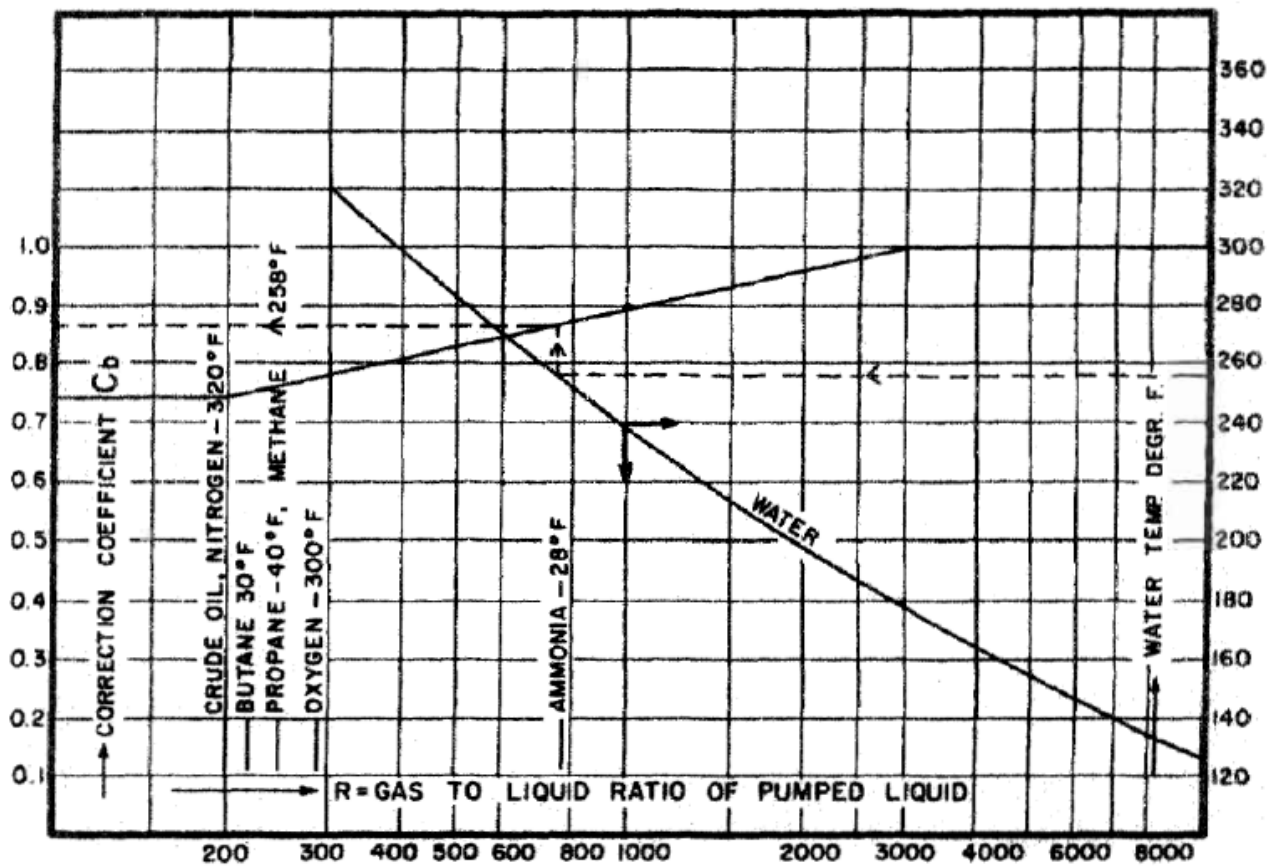

Figure 3. Cb calculation as function of gas to liquid ratio (figure 4 of [1])

Once filled the input data (blank cells of Figure 4), NPSHr40000hr is evaluated according to the application of Vlaming[1] and Shiavello [2] methods.

INPUT DATA		
unit system	SI	
Pump speed	1489	RPM
Qtd	876	m3/h
Htd	310	m
Pump BEP head/stage @max diameter	100.5	m
Pump BEP flow @max diameter:	823.7	m3/h
volumentric efficiency	90.0	%
NPSHav	15.0	
gas to liquid ratio of pumped fluid R	14000.0	
D1 tip - LE	217.0	mm
Dhub - LE	150.0	mm
b1	62.0	mm
beta1-tip	13.0	°
beta1-hub	21.0	°
leading edge thickness s	7.0	mm
length of chamfer of the leading edge L	20.0	mm

double suction pump ☑

Suggested values of R	
for special fluid	
fluid	R
crude oil nitrogen -320°F (-196°C)	200
for water	
Temp	R
50.0	13998.0

Figure 4: Input data for NPSHr40000hr calculation

The curve of NPSHr 40000hr as function of pump flow rate is shown in Figure 8. NPSH available, NPSHr 40000hr curve and operative points are shown.

The evaluation of the NPSHr 40000hr curve is the average value between the Vlaming and Schiavello trade off, shown in Figure 6 and Figure 7, to mitigate the effect of NPSHse worsening at high and low flow, as shown in Figure 5.

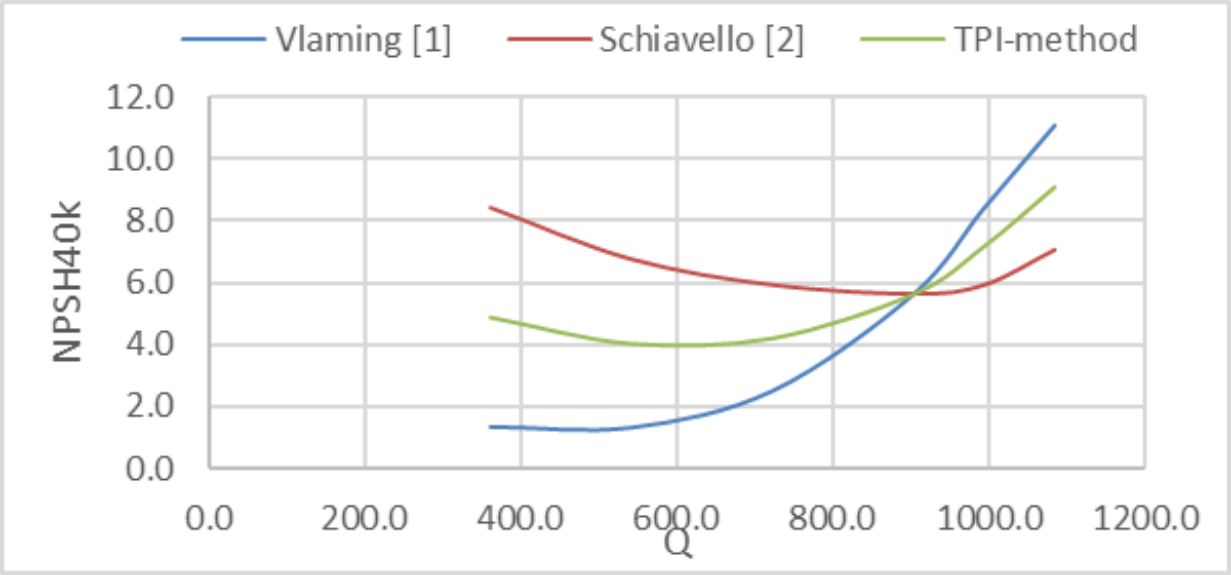

Figure 5.TPI tradeoff for NPSHr40000 versus pump flowrate

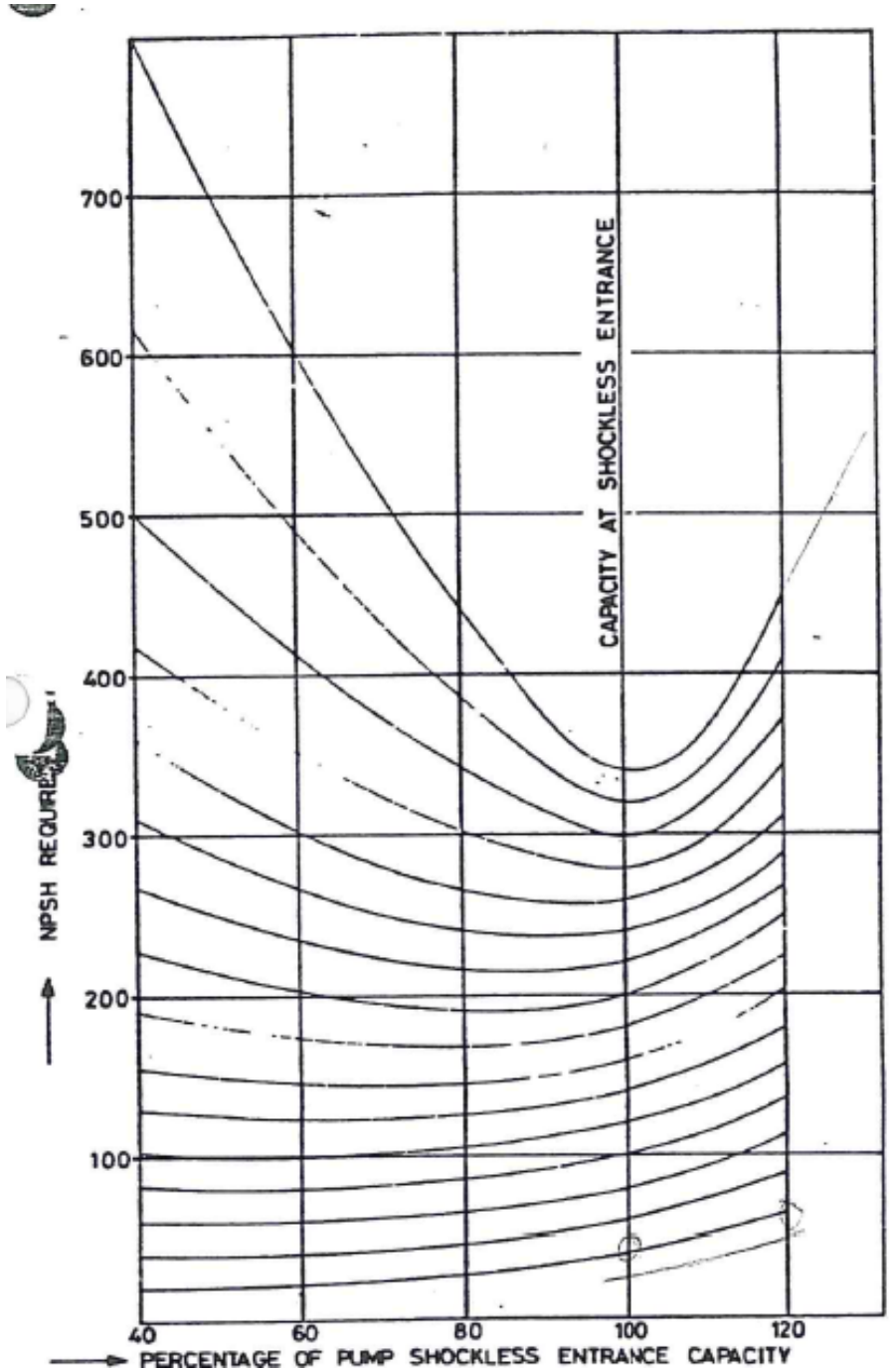

Figure 6. NPSHr40000hr curve as function of pump shockless ratio (figure 5 of [1])

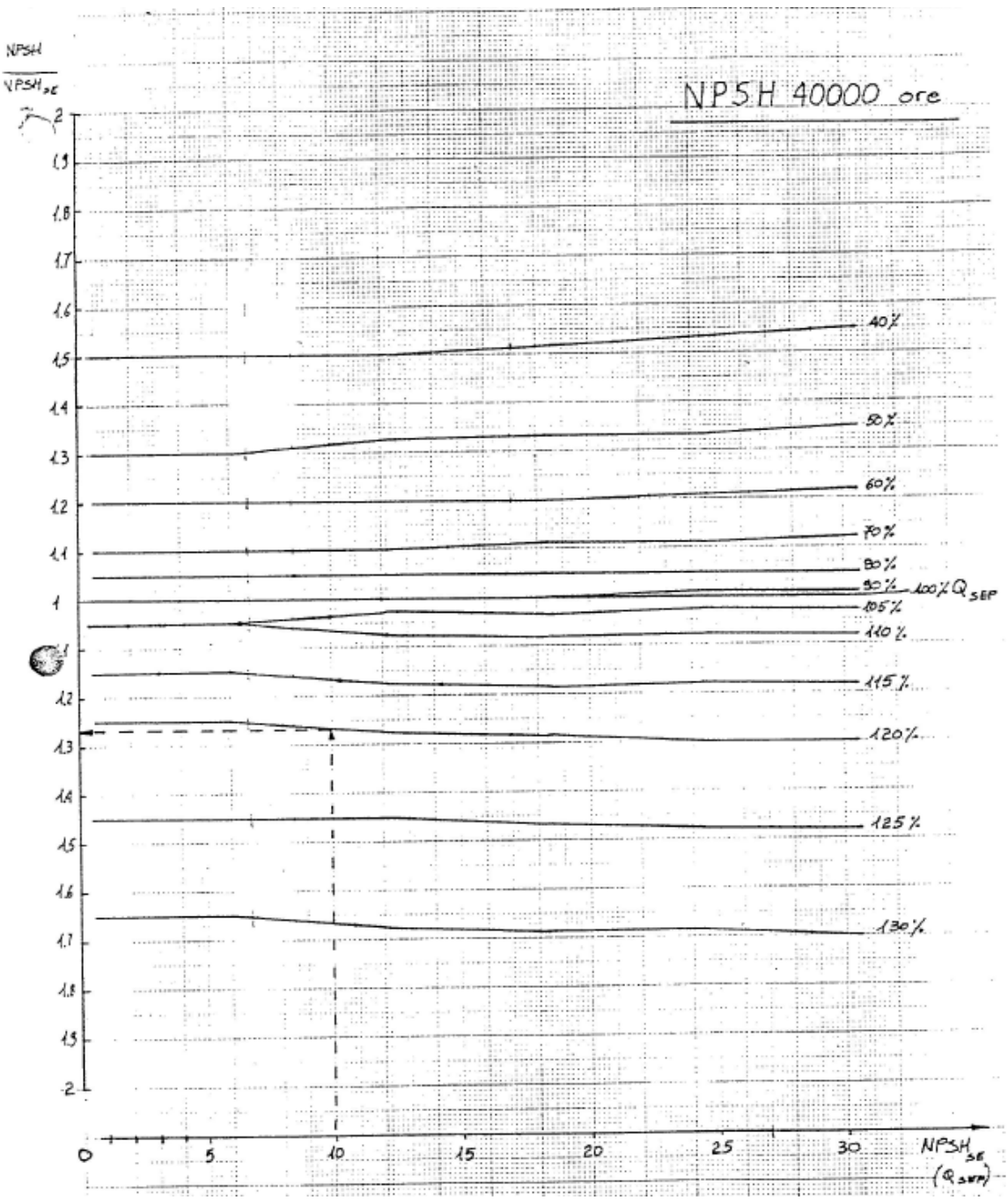

Figure 7.ratio between NPSHR/NPSHRse as function of NPSHse (figure 4 of [2])

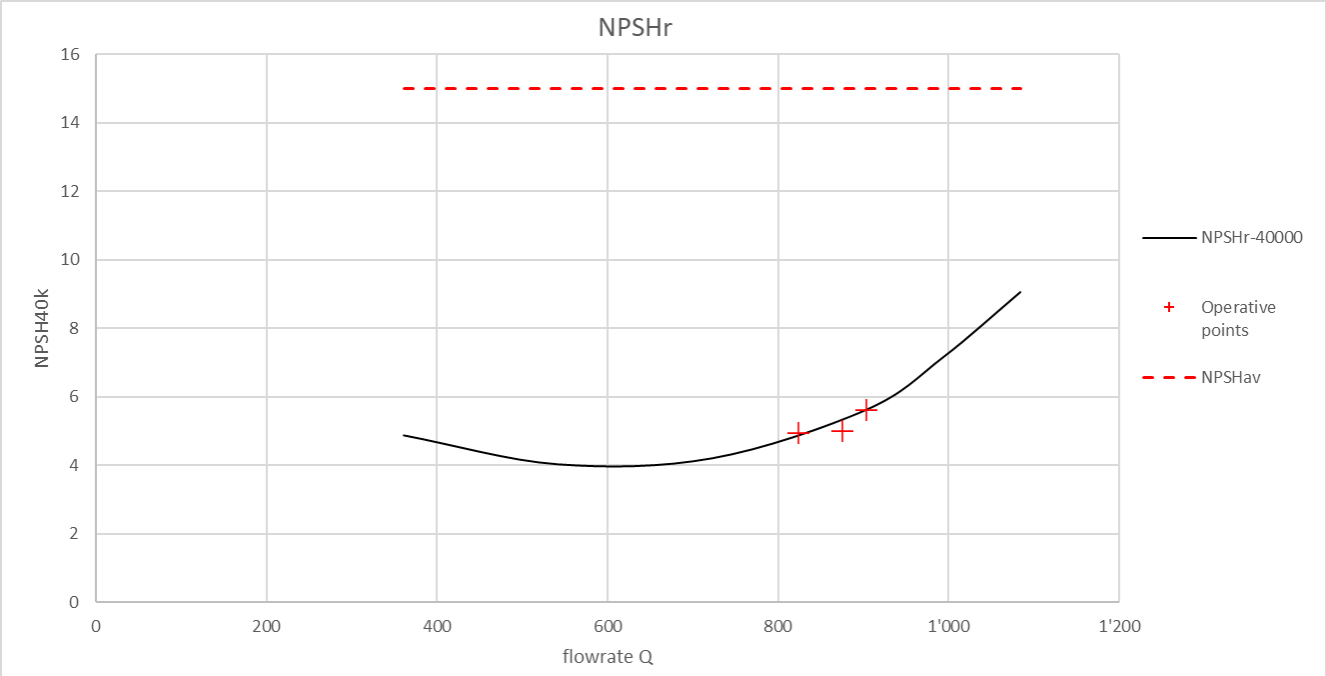

Figure 8: Graph for NPSHr 40000hr-capacity curve

Output data (see Figure 9) are listed below:

- Pump specific speed and power
- Peripheral velocity at tip and hub of the impeller leading edge
- Impeller throat area
- Absolute velocity of the flow at the inlet of the impeller blade on hub and shroud at shockless condition
- Flowrate of the impeller at shockless condition
- Correction factors for specific speed Ca
- Correction factor for gas to liquid radio Cb
- Friction coefficient for the casing k1
- Friction coefficient for the impeller k2
- NPSH40000hr at several operative points (shockless, bep and rated)

OUTPUT DATA

pump specific speed	15.88	Nq
peripheral velocity at D1.tip	16.92	m/s
peripheral velocity at D1.hub	11.69	m/s
impeller inlet area A1	0.0357	m2
cm shockless at blade tip	3.91	m/s
cm.shockless at blade hub	4.49	m/s
impeller flowrate at shockless	1004.06	m3/h
Correction factor for specific speed	1.08	Ca
Correction factor for gas to liquid ratio	1.00	Cb
friction coefficient for pump case	1.20	K1
friction coefficient for impeller	0.28	K2

	pump flowrate	NPSH 40000 hr
	m3/h	m
SHOCKLESS	903.7	5.6
BEP	823.7	4.9
RATED	876	5.0

Figure 9: Output data of NPSHr 40000hr calculation

6. Limitations

The calculation of NPSH40000hr is valid for axial inlet flow only. In case of radial flow approaching the swirl angle contribute must be considered on the calculation of absolute velocity.

7. Acceptance Criteria

The leading edge of the impeller at tip must be tapered to achieve the smallest radius possible, considering mechanical limitations. The length of the taper shall be at least 3 times the thickness of the blade.

The method is not applicable for peripheral speed of the impeller inlet higher than 150ft/s (45m/s) if NPSH40000hr is close to NPSHav, in this situation a physical model test with visual observation of cavitation length for the evaluation of the erosion of the impeller shall be performed.

In case of NPSH3% higher or NPSHav lower than NPSHr 40000 then the selected hydraulic cannot be used for the purpose and the support of Engineering dpt or R&D dpt. is necessary.

A typical comparison of the NPSH curves is shown in Figure 10: the correct selection is achieved when NPSH3% curve is lower than NPSHr40000 and NPSHav is higher than NPSHr 40000 for the entire range of flow rates.

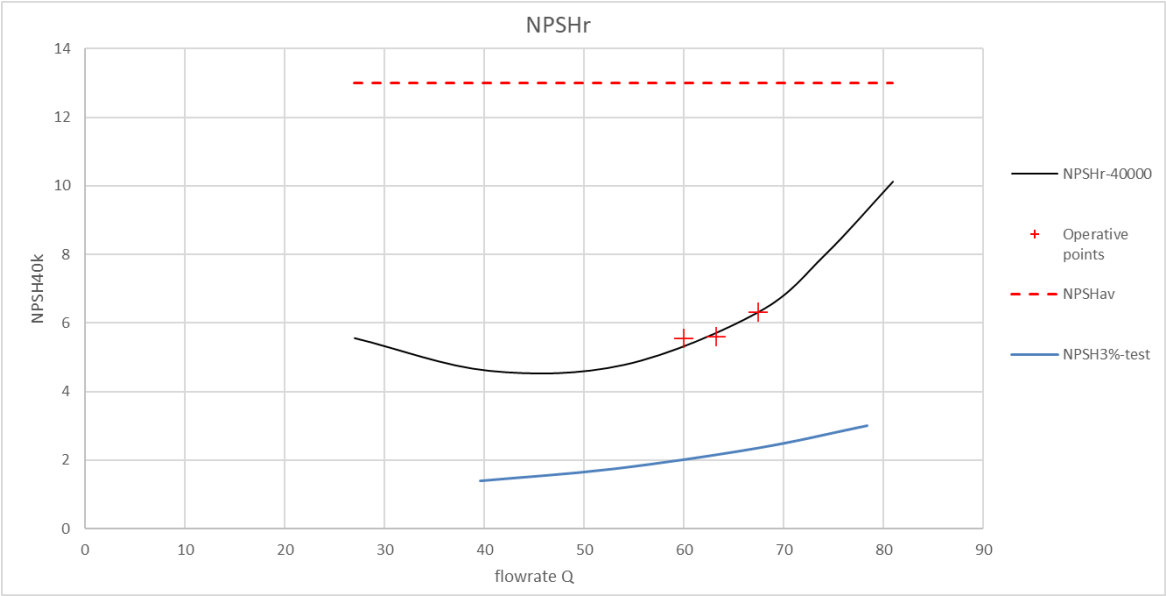

Figure 10. typical NPSHr3% compared to NPSHr40000

8. References

[1] D.J.Vlaming – A method for estimating the net positive suction head required by centrifugal pumps. ASME 81-WA/FE-32

[2] B.Schiavello – Fist stage Impeller Vane Inlet analysis and design